STUDENT PROJECT ASSIGNMENT 2

Title

Experimental Verification of Neutral Point Location for a Wing–Tail Configuration Using Glide Tests

Project Type

Experimental Aerodynamics & Flight Mechanics Project
(Design–Theory–Experiment Integration)

Objective

The objective of this project is to **experimentally verify the location of the neutral point** for a **wing–tail aircraft configuration consisting of two identical wings**, as derived in classroom lectures. The project aims to demonstrate that when **downwash interference is minimal**, the neutral point lies at the **midpoint between the aerodynamic centres of the wing and the tail**, and that **static longitudinal stability depends on the centre of gravity (CG) location relative to this neutral point**.

Background and Motivation

In theoretical flight mechanics, the neutral point represents the boundary between **stable and unstable longitudinal behaviour**. For a configuration consisting of **two identical lifting surfaces (wing and tail)** separated by a distance, analytical derivations predict that the neutral point lies **midway between the aerodynamic centres of the two surfaces**, assuming negligible aerodynamic interference.

This project translates that theoretical derivation into a **real-world experimental demonstration**, allowing students to visually and physically observe stable and unstable flight behaviour by shifting the CG relative to the neutral point.

Experimental Configuration

Geometry and Layout

- Configuration: **Wing–tail combination with identical wings**
- Wing planform: **Rectangular**
- Number of lifting surfaces: **2 (wing + tail)**

- Span of each wing: ~2.4 m
- Chord length: ~0.23 m
- Aspect ratio: ~12
- Separation distance between aerodynamic centers: ~1.76 m

Mass Properties

- Total mass of the configuration: ~1.8 kg
- CG adjustable using **removable ballast weights**
- Aerodynamic centers of wing and tail marked physically on the model

Theoretical Foundation

Students must recall and apply the following theoretical result derived in lectures:

For two identical lifting surfaces with negligible downwash interaction,
Neutral Point (NP) = Midpoint between the aerodynamic centers of the wing and the tail

Static stability condition:

- **Stable:** CG ahead of neutral point
- **Unstable:** CG behind neutral point

Project Tasks

Phase 1 – Theoretical Analysis

Students must:

1. Identify the aerodynamic center of the wing and tail (¼ chord)
2. Measure the separation distance between the two aerodynamic centers
3. Compute the **neutral point location**
4. Predict stable and unstable CG ranges

Phase 2 – Experimental Setup

- Assemble the wing–tail configuration
- Mark:
    a. Aerodynamic centre of wing
    b. Aerodynamic centre of tail
    c. Neutral point (midpoint)
- Verify current CG location using balance tests

Phase 3 – Stable Flight Demonstration

1. Shift CG **slightly ahead of the neutral point** using ballast weights
2. Conduct hand-launched glide tests from:
    a. Ground-level launch
    b. Elevated platform (if permitted and safe)
3. Observe:
    a. Glide smoothness
    b. Pitch behaviour
    c. Flight stability

Phase 4 – Unstable Flight Demonstration

1. Remove ballast and shift CG **behind the neutral point**
2. Repeat glide tests

- Observe:
    a. Pitch instability
    b. Reduced glide performance
    c. Failure to maintain steady flight

Observations and Expected Results

CG Location	Expected Behaviour
Ahead of Neutral Point	Stable glide, smooth pitch response
At Neutral Point	Marginal stability
Behind Neutral Point	Unstable flight, pitch divergence

Students should clearly relate observed behaviour to **static stability theory**.

Deliverables

Each student group must submit:

1. **Project Report** containing:
    a. Introduction and theoretical background
    b. Geometry and mass calculations
    c. Neutral point estimation
    d. Experimental procedure
    e. Results and discussion
    f. Comparison with theoretical predictions
    g. Conclusions
2. **Experimental Evidence**
    - Photos or videos of stable and unstable glide tests

Evaluation Criteria

Component	**Weightage**
Theoretical Analysis	30%
Experimental Setup & Accuracy	25%
Interpretation of Results	25%

Component	Weightage
Report Quality	20%

Learning Outcomes

Upon completion of this project, students will be able to:

- Determine the **neutral point** for a wing–tail configuration
- Relate **CG location to static longitudinal stability**
- Validate analytical derivations using experiments
- Understand the limitations of ideal assumptions (downwash, measurement error)
- Translate flight mechanics theory into physical intuition

Safety Guidelines

- Conduct experiments in open areas only
- Ensure no personnel are in the glide path
- Use elevated launches only with instructor approval
- Handle ballast weights securely

Concluding Remark for Students

This project highlights that **aircraft stability is governed by relative geometry and mass distribution**, not merely by wing shape. The experiment reinforces a fundamental principle of flight mechanics:

Stable flight requires the centre of gravity to be ahead of the neutral point.